

bling, rinse and spin stages. At each stage, the approximate minimum and maximum power consumption may be known, as well as the duration of that stage.

In some circumstances the ESA may allow the stage to be delayed or paused (subject to safety and manufacturer's discretion and user preferences).

Typically, appliances with a heating element cannot have their power consumption adjusted and can only be paused or delayed.

Some ESAs may not be flexible other than a delayed cycle start (for example, once the washing cycle has been started then they run continuously until the cycle completes).

Appliances that only support the PowerForecastReporting and not any of the adjustment features may indicate that they are not flexible in the forecast slot format.

The PowerForecastReporting and the adjustment features aim to align to the [SAREF4ENER] ontology.

Inverter driven ESAs: some inverter driven ESAs can consume or generate a variable amount of power.

For example, a single phase EVSE can be adjusted in the range of 6-32Amps in 0.6 Amp steps in EU or on a hardwired 120V supply in the range of 6-15 Amps in US.

For example, a home battery may be adjusted to charge or discharge in steps of 1W.

For example, a heat pump may be able to modulate its compressor inverter between 20-100% of its rated power.

The ESA indicates its power adjustment range and its nominal power consumption as part of its Forecast.

9.2.4.3. StateForecastReporting Feature

Some ESAs do not know their actual power consumption, but do know the state of operation. Like the PowerForecastingReporting feature, this uses the same slot structure mechanism to indicate a change in state vs time.

An external observing EMS may have access to real-time meter readings, and could learn the typical power consumption based on the advertised internal state of the ESA.

To enable this capability, the ESA SHALL report its internal operational state using an manufacturer specific value.

Once the EMS has built a model of the state vs observed power consumption, it may request a forecast adjustment for particular times of the day, encouraging the ESA to use power at alternative times.

9.2.4.4. StartTimeAdjustment Feature

ESAs which support the Start Time Adjustment feature, allow an EMS to recommend a change to